



Data Analytics for Safety Management

Analytics for Managers - Assessment 2

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Introduction

Modern corporate management relies significantly on data analytics. Organisations can get insightful information regarding consumer behaviour, market trends, and operational effectiveness by utilising big data and sophisticated analytics approaches (De langhe & Puntoni, 2021). This data-driven strategy promotes well-informed decision-making, better resource management, and increased performance across numerous corporate functions (Hume & West, 2020; Boldosova & Luoto, 2020). Data analytics, which range from predictive modelling to real-time monitoring, equips companies with the tools they need to recognize opportunities, manage risks, and optimize their business plans (Boldosova & Luoto, 2020). This boosts their edge and success in the modern, data-rich, and dynamic business environment.

Data-driven decisions in Safety Management

Data-driven decision making has become a staple tactic in many industries, including safety management. Organisations are starting to realise how crucial it is to use data and analytics to enhance safety protocols, lower risks, and create safer working conditions (Laine et al., 2022). By collecting, assessing, and analysing safety-related data, decision-makers can gain meaningful knowledge that inspires proactive safety management practises (Sherratt & Ivory, 2019). Decisions based on data are becoming increasingly crucial in safety management. Managers can mitigate the risk of future mishaps by making educated decisions by gathering and analysing data to find patterns of accidents and incidents (Hume & West, 2020). Safety managers may immediately identify potential hazards and take preventive action before any accident or incident occurs with data-driven decision-making. This enables companies to establish a secure working environment for both their workers and clients.

Data-driven decision making in safety management is the systematic and strategic use of data to find potential threats, trends, and patterns to help organisations prevent accidents, injuries, and other safety events (Yinghui et al., 2021). It provides a more fact-based and objective approach to decision-making by moving away from subjective judgements (De langhe & Puntoni, 2021). Safety professionals can identify areas for improvement, obtain a thorough awareness of their organization's safety performance, and then prioritise resources and efforts in those areas. It enables a proactive and predictive rather than a reactive approach to safety management, helping to avert events before they happen and lessen their effects if they do (Manu et al., 2019).

Obtaining relevant safety data from a range of sources, such as incident reports, safety inspections, near-miss reports, employee feedback, and other data points that provide insights into the safety landscape, is the first step of the process. The data is then investigated using statistical methods, data visualisation tools, and other analytical techniques to spot patterns, trends, and correlations. To track the success of safety measures, key performance indicators (KPIs) such as incident rates, near-miss frequency, safety training completion rates, and compliance with safety protocols are established (Boldosova & Luoto, 2020). These KPIs serve as benchmarks and aspirational objectives. The regular monitoring and reporting of these KPIs enables organisations to evaluate their safety performance, pinpoint areas of concern, and make informed decisions based on data-driven insights.

Data Sets in Data Analytics

To make well-informed business decisions, data sets must be used for data analysis. Large volumes of data can be gathered and analysed to help firms find patterns, trends, and insights that inform strategic decision-making (Boldosova & Luoto, 2020). Data must be cleaned and transformed, statistical methods must be used, and data visualisation tools must be used to properly show results. Businesses can use data analysis to pinpoint client preferences, enhance marketing efforts, boost productivity, and spot potential threats (Laine et al., 2022). Businesses may eliminate uncertainty, make decisions based on solid information, and get a competitive edge in the quickly changing market of today by utilising data-driven insights (Shekari, 2020). It gives businesses the ability to better match their strategies with consumer demands and industry trends, which boosts productivity and spurs expansion.

For this assessment, the following secondary data sets have been identified from Kaggle for running data tests and gaining insights to utilize data records to understand the business issues and derive possible solutions in accordance with the data churning results.

DATA SET 1: OSHA ACCIDENT & INJURY - HSE DATA – 2015-2017

Source: (OSHA Accident and Injury Data, n.d.)

Figure 1: Screenshot of raw data set 1

summary_id	Event Date	Abstract Text	Event Description	Event Keyword	con_end	Construction End Use	build_sto	Building Storie	proj_cost	Project Cost	proj_type
220982664	08/10/2017	At 9:00 a.m. on August 10, EMPLOYEE'S FINGER FINGER,MECHANIC			0		0		0		0
220973937	17/07/2017	At 9:45 a.m. on July 17, 20 EMPLOYEE'S FINGER CAUGHT IN,DRIVE SH			0	Other building	1	1	0		B
220967632	30/06/2017	At 7:30 a.m. on June 30, 20 EMPLOYEE IS HOSPITALAMPUTATED,EXPLC			0		0		0		0
220969901	30/06/2017	At 2:00 p.m. on June 30, 20 EMPLOYEE FALLS FRI RIB,ROOF,HEAD,FA C			0	Commercial building	1	1 D	\$500,000 to \$1, B		
220977748	23/06/2017	At 12:20 p.m. on June 23, EMPLOYEE IS STRUCI STRUCK BY,TRUCK,I			0		0		0		0
220974562	22/06/2017	At 9:07 a.m. on June 22, 20 EMPLOYEE FAINTS V UNCONSCIOUSNES			0		0		0		0
220959936	31/05/2017	At 11:00 p.m. on May 31, 2 FIVE EMPLOYEE KILLI STRUCTURAL COLL			0		0		0		0
220960629	31/05/2017	At 11:00 a.m. on May 31, 2 EMPLOYEE IS STRUCI PIPE,OIL RIG,STRUC			0		0		0		0
220959944	31/05/2017	At 5:15 p.m. on May 31, 20 EMPLOYEE FALLS THI FALL,HEAD,ROOF,F G			0	Sewer/water treatment	2	2 F	\$5,000,000 to \$ B		
220959738	31/05/2017	At 4:45 p.m. on May 31, 20 EMPLOYEE FALLS FO ELEVATED WORK P			0		0		0		0
220960025	31/05/2017	At 4:29 p.m. on May 31, 20 EMPLOYEE IS STRUCI STRUCK BY,DOOR,F			0		0		0		0
220960280	31/05/2017	At 2:30 p.m. on May 31, 20 EMPLOYEE FALLS FRI FALL,LADDER,HEAE B			0	Multi-family dwelling	0		0		A
220959696	30/05/2017	At 12:45 p.m. on May 30, 2 EMPLOYEE IS STRUCI STRUCK BY,MOTORI			0	Highway, road, street	0	A	Under \$50,000 A		
220960082	30/05/2017	At 12:30 p.m. on May 30, 2 EMPLOYEE FALLS FRI FALL,FALL PROTECT G			0	Sewer/water treatment	2	2 E	\$1,000,000 to \$ A		
220968341	30/05/2017	On May 30, 2017, an empl EMPLOYEE IS KILLED CARDIAC ARREST,C			0		0		0		0
220965719	30/05/2017	At 10:01 a.m. on May 30, 20 EMPLOYEE IS STRUCI STRUCK BY,POWER			0		0		0		0
220961437	30/05/2017	At 7:00 p.m. on May 30, 20 EMPLOYEE IS STRUCI STRUCK BY,FALLINC			0		0		0		0
220960041	30/05/2017	At 6:00 p.m. on May 30, 20 EMPLOYEE FOUND U AGRICULTURE,HEA			0		0		0		0
220959605	30/05/2017	At 1:45 p.m. on May 30, 20 EMPLOYEE IS CAUGH CAUGHT IN,ASPHYC			0	Commercial building	1	1	0		B
220960694	29/05/2017	At 10:00 a.m. on May 29, 2 EMPLOYEE STRUCK B TREE TRIMMING,FA			0		0		0		0
220959621	29/05/2017	At 3:00 a.m. on May 29, 20 EMPLOYEE CAUGHT I CAUGHT IN,WIRE R			0		0		0		0
220960462	29/05/2017	At 1:30 p.m. on May 29, 20 EMPLOYEE STRUCK B STRUCK BY,FRACTU			0		0		0		0
220959647	28/05/2017	At 4:15 p.m. on May 28, 20 EMPLOYEE FALLS FRI FALL,ROOF	C		0	Commercial building	1	1 A	Under \$50,000 A		
220960058	27/05/2017	At 1:30 p.m. on May 27, 20 EMPLOYEE TRIMMIN TREE TRIMMING,FA			0		0		0		0
220959126	26/05/2017	At 10:22 a.m. on May 26, 2 EMPLOYEE FALLS FRI FALL,ROOF	C		0	Commercial building	1	1	0		D
220959472	25/05/2017	At 5:00 a.m. on May 25, 20 EMPLOYEE INHALES CHEMICAL VAPOR,			0		0		0		0
220959357	25/05/2017	At 4:00 p.m. on May 25, 20 EMPLOYEE FALLS THI FALL,FLOOR OPENI B			0	Multi-family dwelling	4	4	0		A
220960892	25/05/2017	At 3:15 p.m. on May 25, 20 EMPLOYEE IS KILLED TRENCH,COMBUST M			0	Pipeline	0		0		B
220959340	24/05/2017	At approximately 10:48 or TWO WORKERS ARE SODIUM HYPOCHL			0		0		0		0
220958615	24/05/2017	At 10:00 a.m. on May 24, 2 EMPLOYEE DIES FRO HEART ATTACK	A		0	Single family or duplex	0	A	Under \$50,000 B		
220958953	24/05/2017	At 6:50 a.m. on May 24, 20 EMPLOYEE DIES FRO HEART ATTACK			0		0		0		0
220959241	24/05/2017	At 2:43 p.m. on May 24, 20 EMPLOYEE STRUCK B TREE TRIMMING,ST			0		0		0		0
220958490	24/05/2017	At 1:45 p.m. on May 24, 20 EMPLOYEE DIES FRO HEART ATTACK,HE/			0		0		0		0

This OSHA data collection includes elements like workplace incident reports, injury and sickness statistics, safety infractions, and inspection records. It also includes details about particular workplace dangers or infractions as well as information about businesses, industries, and places.

DATA SET 2: WORK-RELATED INJURY RECORDS – YOY 2016-2021

Source: (*Work Related Injury Data (2016-2021)*, n.d.)

Figure 2: Screenshot of raw data set 2

ID	Company_Name	Establishment_Name	ein	Street_Address	City	State	Zip Code	naics_code	Industry_Desc	Annual_Avg_Employee	Total_Hrs_Worked
892	Genesis Healthcare	Milford Center		10 Veterans Memorial Drive	Milford	MA	1757	623110	Nursing homes	108	216077
5713	Genesis Healthcare	Langdon Place of Dover		60 Middle Road	Dover	NH	3820	623110	Nursing homes	112	223000
6075	genesis healthcare	bennington health and rehab		2 blackberry lane	bennington	VT	5201	623110	Nursing homes	95	190035
6156	Genesis HealthCare	Coventry Center		10 Woodland Drive	Coventry	RI	2816	623110	Nursing homes	206	411604
6500	Genesis HealthCare	The Greenwood Center		1139 Main Ave	Warwick	RI	2886	623110	Nursing homes	125	125972
6504	Genesis Healthcare	Sunset Point Genesis Healthcare		1980 Sunset Point Rd.	Clearwater	FL	33765	623110	Skilled nursing fac	110	219672
6967	Genesis Healthcare	Langdon Place of Nashua		319 E. Dunstable Rd	Nashua	NH	3062	623311	Assisted-living fac	67	134187
7214	Genesis Healthcare	Sarah Brayton Nursing Center		4901 North Main Street	Fall River	MA	2720	623110	Nursing homes	211	422794
7447	Genesis HealthCare	Academy Manor		89 Morton Street	Andover	MA	1810	623110	Nursing homes	185	370713
7615	Genesis Healthcare	Wolfeboro Bay Center		39 Clipper Drive	Wolfeboro	NH	3894	623110	Nursing homes	97	194964
7822	Genesis Healthcare	Cedar Ridge Center		23 Cedar Ridge Drive	Skowhegan	ME	4976	623110	Nursing homes	97	194600
8160	Genesis Healthcare	Sandy River Center		119 Livermore Falls Road	Farmington	ME	4938	623110	Nursing homes	103	179657
8204	GENESIS HEALTHCARE	Village Green of Bristol		23 Fair Street	Forestville	CT	6010	623110	Nursing homes	112	229495
8529	Genesis Healthcare	Apple Valley Center		400 Groton Road	Ayer	MA	1432	623110	Nursing homes	125	249757
8700	Genesis HealthCare	Prescott House		140 Prescott Street	North Andov	MA	1845	623110	Nursing homes	143	285525
8766	Genesis Healthcare	Huntington Place		1775 Huntington Ln.	Rockledge	FL	32955	623110	Nursing homes	89	177306
9157	Genesis HealthCare	Windward Gardens		105 Mechanic St.	Camden	ME	4843	623110	Nursing homes	78	155748
9705	Genesis HealthCare	RiverRidge Center		3 Brazier Lane	Kennebunk	ME	4043	623110	Nursing homes	126	252308
11652	Genesis Healthcare	Cottage of the Shoals		500 John Aldridge Drive	Tuscumbia	AL	35674	623110	Nursing homes	98	195851
12093	Genesis Healthcare	Orchard Ridge Center		49274 Voorhees Rd	New Port Rici	FL	34653	623110	Hospices, inpatient	118	235807
12220	Genesis Healthcare	Mountain View Center		9 Haywood Ave	Rutland	VT	5701	623110	Skilled nursing fac	162	323594
12308	Genesis HealthCare	Kent Regency		660 Commonwealth Avenue	Warwick	RI	2886	623110	Nursing homes	189	377981
12326	Genesis HealthCare	Exeter Center		8 Hampton Rd	Exeter	NH	3833	623110	Nursing homes	70	145600
12328	Genesis Healthcare	Westford House		3 Park Drive	Westford	MA	1886	623110	Nursing homes	139	277746
12331	Genesis Healthcare	Oak Grove Center		27 Cool Street	Waterville	ME	4901	623110	Nursing homes	115	229711
12344	Genesis Healthcare	Ridgewood Center NH		25 Ridgewood Rd	Bedford	NH	3110	623110	Nursing homes	168	335067
12353	Genesis Healthcare	Rochester Manor		40 Whitehall Rd	Rochester	NH	3867	623110	Nursing homes	93	190469
12361	Genesis Healthcare	Heritage Nursing Care Center		841 Merrimack St	Lowell	MA	1854	623110	Nursing homes	138	276835
12406	Genesis Healthcare	Quindley Center		888 North Main St	Andover	MA	3301	623110	Nursing homes	110	220104

Workplace injury data includes characteristics like the type and nature of the injury (for example, fractures, sprains, or burns), the body parts affected, the severity of the injury, the demographics of the incident, the injured worker's industry or occupation, the scene of the incident, and factors that contributed to the injury (for example, slips, falls, or machinery accidents).

Understanding the data variables: Identifying and examining the various types of data that are contained in a dataset is necessary to comprehend data variables (Yinghui et al., 2021). To acquire insights and carry out the necessary statistical analyses, this includes recognising categorical factors (such as gender, product category, and age) and numerical variables (such as sales numbers, and sales figures).

Figure 3: Data Set 1 - Data elements & definitions.

Data Element	Data Element Definition	Variable Type
summary_nr	Unique number on the event record	Nominal
Event Date	Date of the event	Interval/ Ratio
Abstract Text	Text record of the reported accident	Nominal
Event Description	Event described in OSHA description	Nominal
Event Keywords	Keywords from the event description	Nominal
con_end	Coded - type of construction	Nominal
Construction End Use	Type of construction	Nominal
build_stor	Coded - number of stories in the building	Interval/ Ratio
Building Stories	Number of stories in the building	Interval/ Ratio
proj_cost	Coded range - cost of the project	Interval/ Ratio
Project Cost	Bucketed cost of the project	Interval/ Ratio
proj_type	Coded - Type of project	Nominal
Project Type	Type of project undertaken (New, Alteration, Demolition etc	Nominal
Degree of Injury	Type of injury (Fatal or Non Fatal)	Nominal
nature_of_inj	Coded - list of injury types	Nominal
Nature of Injury	Type of injury	Nominal
part_of_body	Coded - part of body impacted	Nominal
Part of Body	Part of body impacted	Nominal
event_type	Coded - description of event type	Nominal

Figure 4: Data Set 2 - Data elements & definitions.

Data Element	Data Element Definition	Variable Type
ID	Unique number for each record	Nominal
establishment_name	The name of the establishment reporting data.	Nominal
ein	Employer Identification Number (EIN) is also known as Federal Tax Identification Number	Nominal
company_name	The name of the company that owns the establishment.	Nominal
street_address	The street address of the establishment.	Nominal
city	The city where the establishment is located.	Nominal
state	The state where the establishment is located.	Nominal
zip_code	The full zip code for the establishment.	Nominal
naics_code	The North American Industry Classification System (NAICS) code which classifies an establishment's business.	Nominal
industry_description	Industry Description	Nominal
size	The size of the establishment based on the maximum number of employees that worked there at any point in the year you are submitting data for. Enter 1 if the establishment has < 20 employees Enter 2 if the establishment has 20-249 employees Enter 3 if the establishment has 250-999 employees Enter 4 if the establishment has 1000-9999 employees Enter 5 if the establishment has 10000 or more employees	Ordinal
year	Year of data collection	Interval/Ratio
area	Area of data collection	Nominal
to	to	
to	to	
total_djtr_cases	Total number of cases with job transfer or restriction (Form 300A Field I) Must be a number	Interval/Ratio
total_other_cases	Total number of other recordable cases (Form 300A Field J)	Interval/Ratio
total_dafw_days	Total number of days away from work (Form 300A Field K)	Interval/Ratio
total_djtr_days	Total number of days of job transfer or restriction (Form 300A Field L)	Interval/Ratio
total_injuries	Total number of injuries (Form 300A Field M(1)) Must be >= 0 Must be a number	Interval/Ratio
total_skin_disorders	Total number of skin disorders (Form 300A Field M(2))	Interval/Ratio
total_respiratory_condit	Total number of respiratory conditions (Form 300A Field M(3))	Interval/Ratio
total_poisonings	Total number of poisonings (Form 300A Field M(4))	Interval/Ratio
total_hearing_loss	Total number of hearing loss (Form 300A Field M(5))	Interval/Ratio
total_other_illnesses	Total number of all other illnesses (Form 300A Field M(6))	Interval/Ratio
establishment_ID	Unique number for each establishment created under an individual user account	Nominal
created_timestamp	The date and time a record was submitted to the ITA	Interval/Ratio

Data Problems

Data can be helpful in finding solutions to business issues. Businesses can find hidden patterns, trends, and correlations in their data by utilising data analytics tools (Betsis et al., 2019). They can then make data-driven decisions and take preventative measures to deal with problems. Understanding client behaviour, optimising resource allocation, discovering operational inefficiencies, and projecting market trends are all made possible with the use of data analysis (Wang et al., 2021). It enables companies to reduce risks, enhance customer happiness, and improve product or service offerings (Betsis et al., 2019). Businesses can acquire a competitive edge, spur innovation, and achieve sustainable growth in today's dynamic and data-centric corporate environment by using data to guide decision-making processes.

For this assessment, the following data problems have been identified to be potentially solved using data analysis techniques, data interpretation and utilization of data testing procedures.

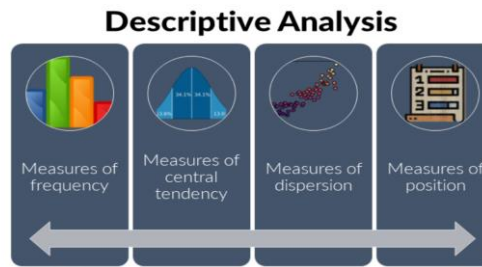
- ◆ **Identifying and analysing factors causing most accidents for designing appropriate training mediums.**
- ◆ **Ascertaining the highly impacted areas basis the data records and forecasting future incident trends.**
- ◆ **Gaining deeper insights into various factors of the accident data records and establishing the influence of the factors on the degree of injury caused.**

Data-driven techniques and types of analysis

Descriptive analysis

In safety management, descriptive data analysis is highly significant. To find trends, patterns, and insights regarding incidents, injuries, and near-misses, historical safety data must be analysed. Organisations can better evaluate safety performance, identify high-risk areas, and prioritise changes by using tools like data visualisation, statistical analysis, and data mining approaches (Boldosova & Luoto, 2020). Descriptive analysis supports continuous development in safety management practises by monitoring safety metrics, locating incident causes, assessing the success of safety efforts, and measuring performance against industry norms.

Figure 5: Depiction of Descriptive Data Analysis



Diagnostic analysis

In safety management, diagnostic data analysis involves looking at prior information to pinpoint the underlying causes and elements that contributed to safety events or near misses. Techniques including trend analysis, correlation analysis, root cause analysis, and fault tree analysis are frequently used in this analysis (Hume & West, 2020). Organisations can learn more about the underlying causes of safety accidents, identify common risk factors, and take corrective action to avoid recurrences by analysing patterns and linkages in the data. Understanding the causes of harmful conditions or behaviours through diagnostic data analysis enables companies to conduct targeted safety interventions and enhance overall safety performance.

Figure 6: Depiction of Diagnostic Data Analysis

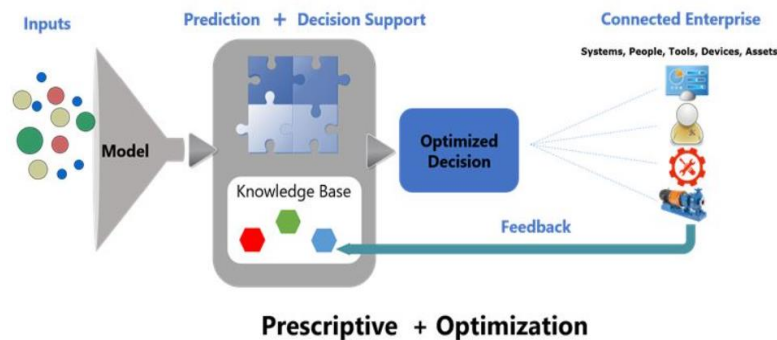


Prescriptive analysis

Prescriptive data analysis for safety management uses cutting-edge analytics approaches to examine past safety data, spot trends and risk factors, and offer practical advice for averting mishaps and enhancing safety procedures (Shekari, 2020). To provide a safer working environment, it aids organisations in proactively addressing safety issues, putting forth preventive measures, and enhancing safety procedures. Decision trees, optimisation models, simulation techniques, and machine learning algorithms are examples of prescriptive data analysis tools for safety management (Lee et al., 2022). These technologies give businesses the ability to anticipate future safety risks, pinpoint the best safety procedures, properly

allocate resources, and simulate scenarios to take preventative action and enhance overall safety performance.

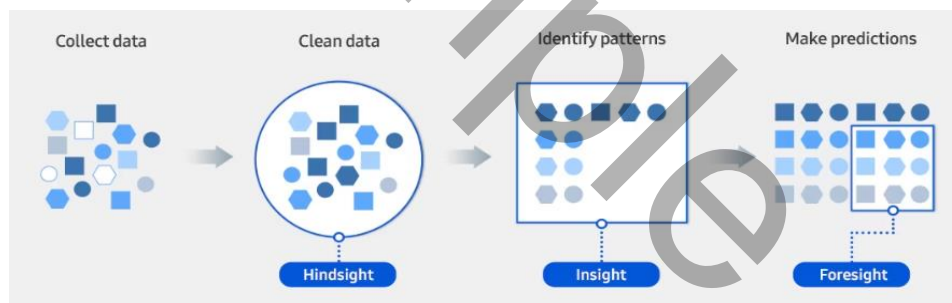
Figure 7: Depiction of Prescriptive Data Analysis



Predictive analysis

Utilising past safety data and statistical models, predictive data analysis in safety management entails making forecasts and predictions about upcoming safety consequences (Boldosova & Luoto, 2020). It seeks to foresee safety problems, proactively identify potential hazards, and put preventative measures in place. Organisations can identify hidden patterns, correlations, and risk factors that may not be visible through descriptive research alone by utilising sophisticated analytics and machine learning approaches (Lee et al., 2022).

Figure 8: Depiction of Predictive Data Analysis



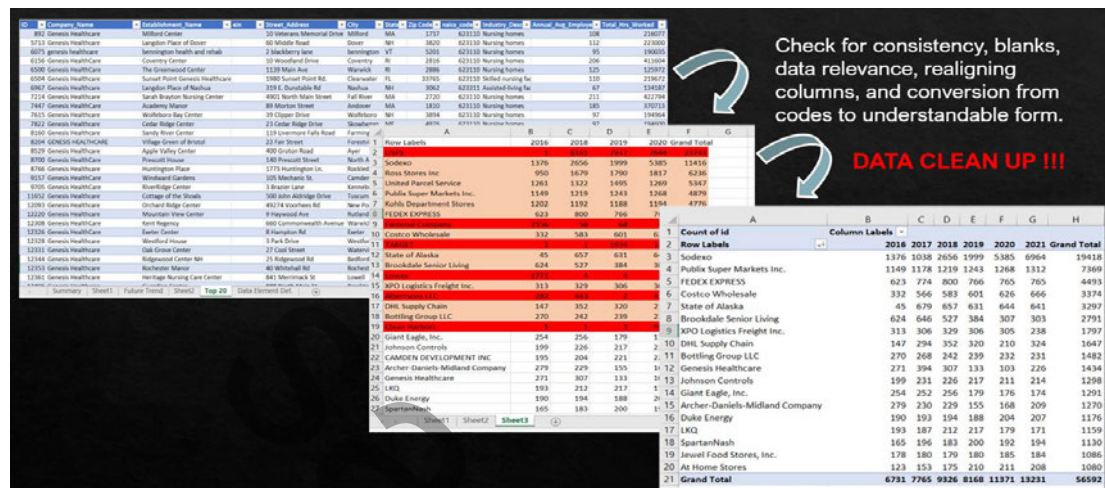
Data Cleaning Process

Data cleaning is crucial when analysing data for safety management. To assure the data's correctness and dependability, it entails locating and fixing flaws, inconsistencies, and abnormalities in the data. In the context of safety management, clean data is essential since it serves as the basis for making wise decisions and putting into action practical safety measures (De langhe & Puntoni, 2021). Data cleaning is beneficial in several ways for safety management. The risk of making decisions based on inaccurate or deceptive data is reduced in the first place since it assures that the analysis is founded on trustworthy and reliable information. The accuracy of safety measures or indicators may be impacted by

inconsistencies or missing numbers, which can be found and fixed with the use of data cleaning (Yinghui et al., 2021).

In brief, data cleaning helps organisations proactively manage risks, enhance safety procedures, and create a safer work environment for employees. It also enables the creation of actionable insights and facilitates evidence-based decision-making in safety management.

Figure 9: Data clean-up & drill down for Data Set 2 – Selecting Top 20 Companies for analysis.

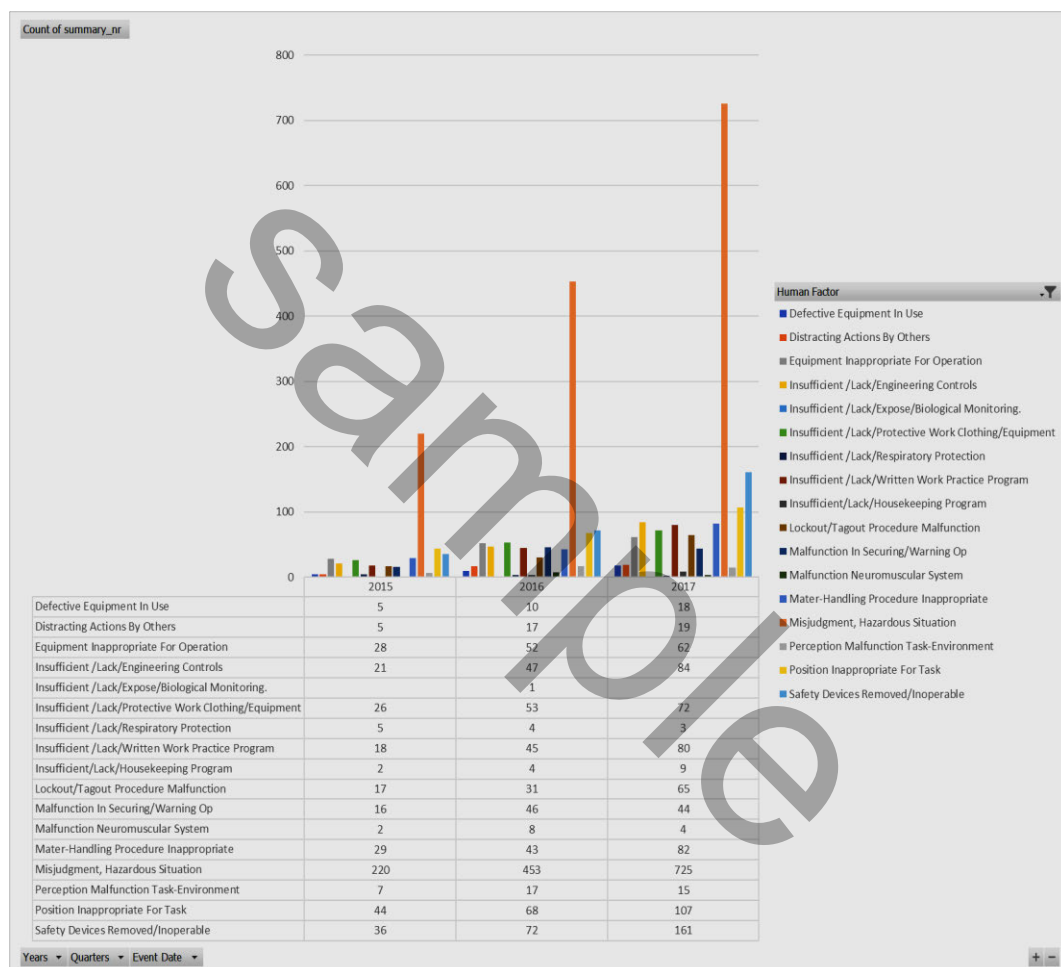


Analysis of Data sets for solutions to Data problems

Analysis for Data Problem 1

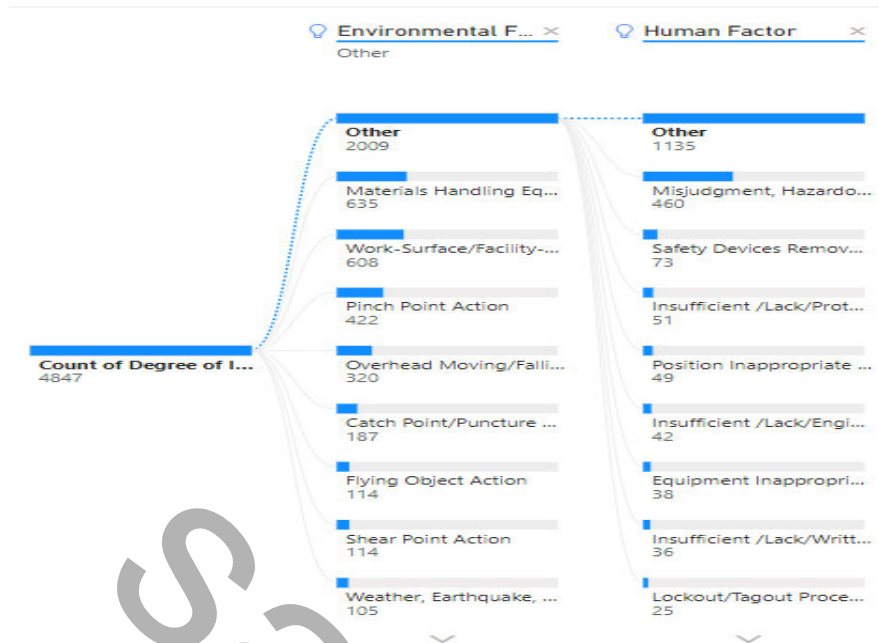
Identifying and analysing factors causing most accidents for designing appropriate training mediums.

Figure 10: Graphical representation of data records basis Human Factor



Accidents that are the result of human factors are those in which human actions or behaviours cause accidents or safety problems to occur. These elements can include mistakes, failures in judgement, a lack of knowledge or awareness, exhaustion, distraction, and disregard for safety procedures. Cerezo-Narváez et al., (2019) highlights organisations can identify the root causes of incidents, implement targeted safety interventions, improve training programmes, and promote a safety culture that emphasises human performance and error prevention by understanding the causes of accidents and analysing accidents from a human factors' perspective (De langhe & Puntoni, 2021).

Figure 11: Level 2 drill down by adding the environmental factor element



Workplace misjudgements by humans can lead to accidents in a variety of settings. Miscalculation of loads or distances during construction might result in falls or structural breakdowns. In construction, improper machine use or a disregard for instructions can lead to accidents or equipment damage (Betsis et al., 2019). Accidents in the transportation industry might result from misjudging speed, distance, or traffic conditions. These mishaps demonstrate the value of safety culture development, risk assessment, training, and clear communication in reducing the effects of human error in various work situations (Sherratt & Ivory, 2019). Occupational safety and hazard data is essential for organizations to identify the training needs of their employees. By leveraging this data, employers can ensure that their workers are adequately trained in safety procedures and protocols to minimize the risks of workplace accidents and injuries.

Figure 12: Level 3 drill down by plotting final results as keywords in a Word Cloud form for designing trainings



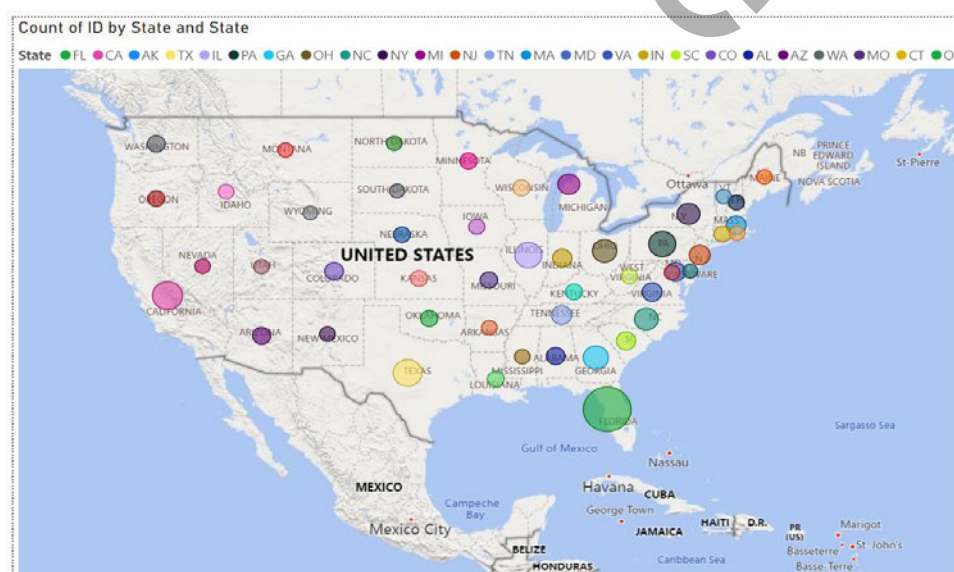
There are several important aspects to be considered when developing training programmes that can effectively reduce workplace accidents brought on by human error and poor judgement in diverse environmental circumstances. These include developing accurate simulations, combining practical exercises, giving constant reinforcement and practise, emphasising situational awareness, encouraging good communication and teamwork, and incorporating risk assessment and decision-making techniques (Betsis et al., 2019). To ensure that trainees acquire the skills, knowledge, and judgement needed to prevent accidents and make wise decisions in their respective work environments, training must be specifically tailored to environmental factors such as hazardous materials, extreme weather conditions, or high-risk work areas.

Analysis for Data Problem 2

Ascertaining the highly impacted areas basis the data records and forecasting future incident trends.

Workplace accidents in the United States demonstrated a worrying trend between 2016 and 2021. The data set processed shows that there was an overall rise in work-related accidents and fatalities over this time. Accidents were frequently caused by falls, transportation mishaps, collisions with tools and items, and exposure to toxic substances. Accident rates were greater in some industries, including manufacturing, transportation, and construction. There have been continual initiatives to increase worker safety, including tighter legislation, training courses, and awareness campaigns. To safeguard the health and safety of American workers, however, continual attention and preventive actions are required given the frequent frequency of workplace accidents.

Figure 13: Geo map radar of most incident records across the United States



[Extract from Power BI – summary : At 7,461,FL had the highest Count of ID and was 746,000.00% higher than VI, which had the lowest Count of ID at 1.] FL accounted for 13.18% of Count of ID. Across all 55 State, Count of ID ranged from 1 to 7,461. (Explained in solutions)]

The forecast of future work accident trends in the United States is tricky using data from 2016 to 2021. However, it is anticipated that continuous attention to enhancing safety legislation, training, and technology adoption will result in a gradual drop in work-related accidents and a stronger emphasis on risk management and prevention strategies.

Figure 14: Overall predictive trend for next 3 years basis data records (2016-2021)

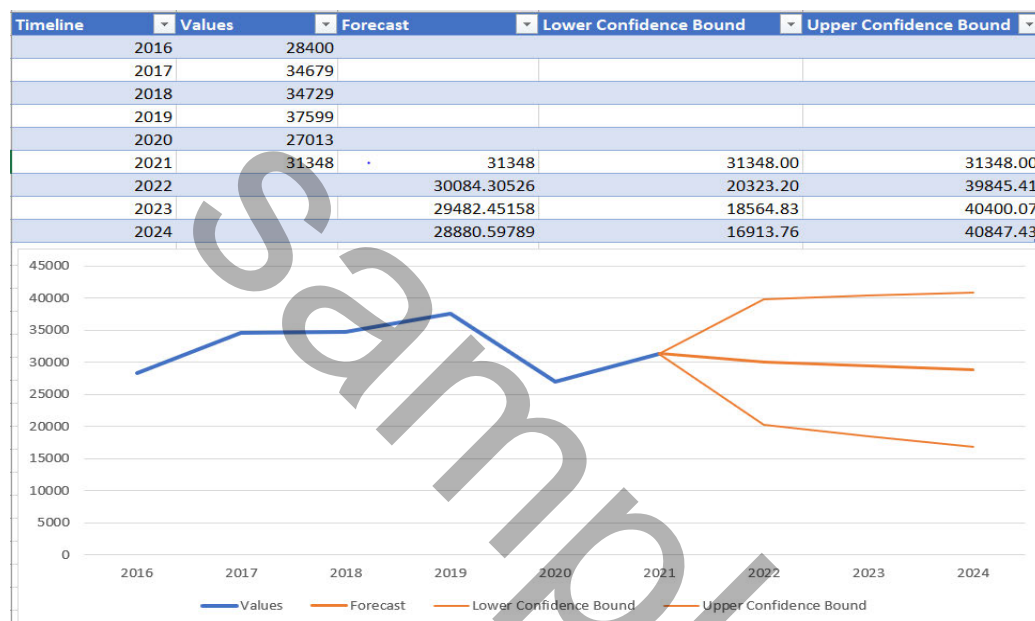
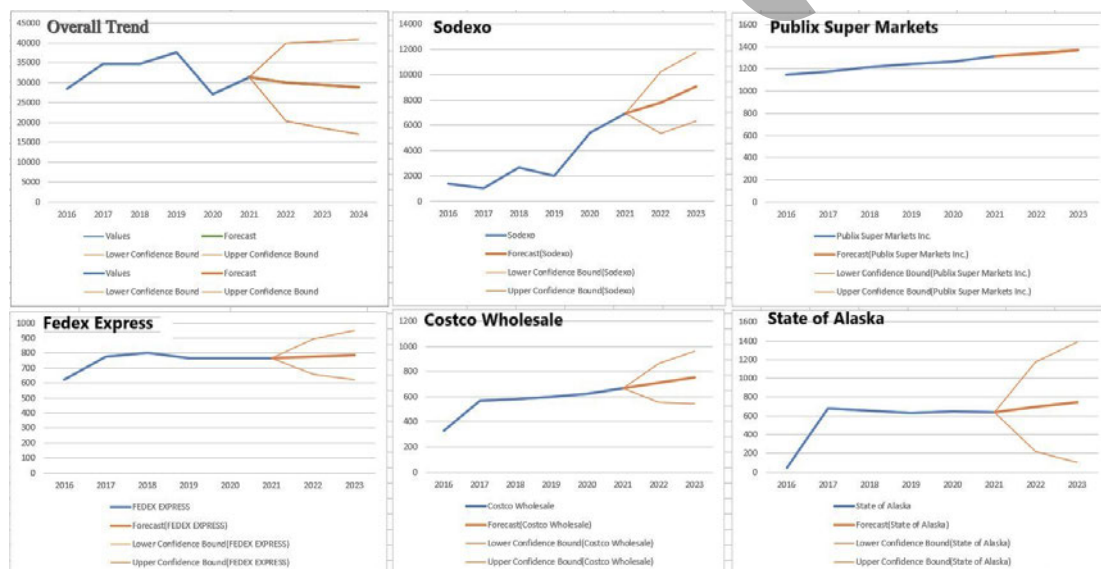


Figure 15: 3-year forecast for Top 5 Entities vs Overall Trend



Analysis for Data Problem 3

Gaining deeper insights into various factors of the accident data records and establishing the influence of the factors on the degree of injury caused.

Correlations

Attributes	Degree ...	event_t...	evn_fac...	nature_...	task_as...
Degree of...	1	0.145	0.285	0.421	0.077
event_type	0.145	1	0.354	0.152	0.020
evn_factor	0.285	0.354	1	0.343	0.019
nature_of...	0.421	0.152	0.343	1	0.065
task_assi...	0.077	0.020	0.019	0.065	1

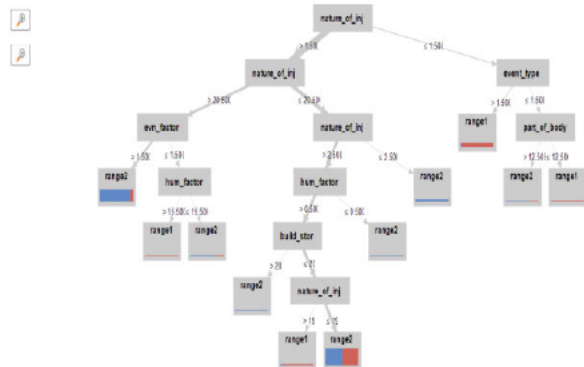
Weights by Correlation

Attribute	Weight
nature_of_inj	0.421
evn_factor	0.285
event_type	0.145
task_assigned	0.077
part_of_body	0.076
hum_factor	0.062
build_stor	0.033

Understanding the underlying trends and enhancing safety measures requires a thorough examination of the relationships between the kind of injury, environmental factors, and event types in workplace accidents. It is feasible to find probable links by looking at the data from work accidents (Betsis et al., 2019). For instance, working in elevated settings may increase your risk of falling from a height. Like this, certain injury categories like burns or slips/trips may be related to environmental conditions like high temperatures or slick surfaces. This study can help in the development of focused treatments, such as bettering working at heights training or putting environmental hazard mitigation measures in place, to lessen the incidence of particular injury types linked to specific environmental factors and event types (Wang et al., 2021). Workplace accidents are a serious concern for any business. It is essential to understand the factors that contribute to workplace accidents to reduce them. One of these factors is hazard exposure, which can be defined as the amount of risk present in an environment (Sherratt & Ivory, 2019). By correlating work accidents with hazard exposure, businesses can benefit from analysing safety management to minimize accidents and other hazards in the workplace. Through examining the processes, they could gain useful insights on how to make their operations safer.

Figure 16: Decision tree on top 3 factors on degree of injury

Figure 17: Decision Tree lift chart - Rapidminer



Decision Tree - Lift Chart

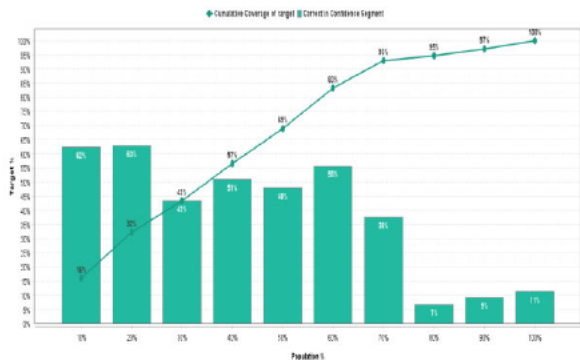


Figure 18: Further drill down on the Top 3 factors to establish the influence.

SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.83039902							
R Square	0.68956254							
Adjusted R Square	0.68897109							
Standard Error	0.43601706							
Observations	4847							
<i>ANOVA</i>								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>gnificance F</i>			
Regression	7	2043.86	291.98	1535.84	0			
Residual	4840	920.137	0.19011					
Total	4847	2964						
	<i>Coefficients</i>	<i>tandard Err</i>	<i>t Stat</i>	<i>P-value</i>	<i>lower 95%</i>	<i>Upper 95%</i>	<i>lower 95.0%</i>	<i>upper 95.0%</i>
Intercept	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
nature_of_inj	0.0238186	0.00087	27.3771	2E-153	0.02211	0.02552	0.02211	0.02552
event_type	0.00434862	0.00148	2.94699	0.00322	0.00146	0.00724	0.00146	0.00724
evn_factor	0.01286433	0.0011	11.6592	5.3E-31	0.0107	0.01503	0.0107	0.01503
task_assigned	0.09070673	0.00953	9.51965	2.7E-21	0.07203	0.10939	0.07203	0.10939
build_star	0.00453172	0.00208	2.17552	0.02984	0.00045	0.00862	0.00045	0.00862

- ◇ Human error factors cause most accidents, future training modules to be made the same and work-specific training to be ensured with highlights on avoiding the seven type causing accidents.
- ◇ From all data records, Florida had the highest accidents. Focusing on the top bucket locations basis the across all 55 State, Count of ID ranged from 1 to 7,461. Basically targeting improvements at the high impact location will give out insights to be horizontally deployed at places with lower numbers, also helping in restricting the increase in the numbers.
- ◇ Most influential factors contributing to accidents and the ones to be focused upon to ensure a downward future trend.
 - ◇ Nature of Work
 - ◇ Event Type
 - ◇ Environmental Factor
 - ◇ Affecting the Degree of Injury

Recommendations and Conclusion

The following recommendations can be put into practise to improve safety management and reduce workplace accidents brought on by human factors, environmental variables, and bad judgements:

- To find potential safety issues, carry out thorough hazard evaluations and risk assessments.
- Create specialised training programmes that address certain environmental concerns, human aspect issues, and decision-making abilities.
- To lessen environmental risks and rely less on individual judgement, implement engineering controls and safety procedures.
- Continued safety practises education and reinforcement are provided.
- Based on the knowledge gained from mishaps and near-misses, review and update safety policies and procedures on a regular basis.
- Encourage staff to quickly identify and report any potential hazards or dangers to promote a proactive approach to safety.

In conclusion, managing workplace dangers and guaranteeing work safety depend on good safety management. Organisations can safeguard their workers, lower workplace accidents, and foster a positive culture by putting safety first. This calls for an all-encompassing strategy that considers risk assessments, thorough training programmes, ongoing oversight, and proactive steps to address human factors, environmental concerns, and errors in judgement. To adjust to changing conditions, safety policies and procedures must be regularly evaluated and improved (Wang et al., 2021). In the end, a dedication to safety management not only protects the health and lives of employees but also generates productivity, morale, and a secure and productive work environment.

Data analytics can be used to improve effective safety management. Organisations can detect high-risk locations and activities, obtain insightful information about safety patterns, and create focused responses by utilising data analytics tools (De langhe & Puntoni, 2021). With the aid of data analytics, patterns and correlations in safety data can be found, enabling proactive decision-making and risk-reduction tactics. It aids in identifying possible safety risks, allocating resources optimally, and gauging the success of safety initiatives (Yinghui et al., 2021). Organisations can enhance safety performance, stop accidents, and make their workplaces safer for their employees by utilising the power of data analytics.

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References

- Betsis, S., Kalogirou, M., & Pertziniidou, M. (2019). Work Accidents Correlation Analysis for Construction Projects in Northern Greece 2003–2007: A Retrospective Study. *Safety*, 5(2), 33. <https://doi.org/10.3390/safety5020033>
- Boldosova, V., & Luoto, S. (2020). Storytelling, business analytics and big data interpretation: Literature review and theoretical propositions. *Management Research Review: MRN*, 43(2), 204–222. <https://doi.org/10.1108/MRR-03-2019-0106>
- Cerezo-Narváez, A., Link to external site, this link will open in a new window, Córdoba-Roldán, A., Pastor-Fernández, A., Aguayo-González, F., Otero-Mateo, M., Link to external site, this link will open in a new window, Ballesteros-Pérez, P., & Link to external site, this link will open in a new window. (2019). Training Competences in Industrial Risk Prevention with Lego® Serious Play®: A Case Study. *Safety*, 5(4), 81. <https://doi.org/10.3390/safety5040081>
- De langhe, B., & Puntoni, S. (2021). Leading With Decision-Driven Data Analytics. *MIT Sloan Management Review*, 62(3), 1–4.
- Hume, E., & West, A. (2020). Becoming a Data-Driven Decision Making Organization. *The CPA Journal*, 90(4), 32–35.
- Laine, J., Delclos, G., Emery, R., & Peskin, M. (2022). Safety & Health Beyond the Workplace. *Professional Safety*, 67(7), 37–38.
- Lee, C. S., Cheang, P. Y. S., & Moslehpour, M. (2022). Predictive Analytics in Business Analytics: Decision Tree. *Advances in Decision Sciences*, 26(1), 1–29.
- Manu, P., Link to external site, this link will open in a new window, Poghosyan, A., Mahamadu, A.-M., Mahdjoubi, L., Gibb, A., Behm, M., & Akinade, O. O. (2019). Design for occupational safety and health: Key attributes for organisational capability. *Engineering, Construction and Architectural Management*, 26(11), 2614–2636. <https://doi.org/10.1108/ECAM-09-2018-0389>

OSHA Accident and Injury Data. (n.d.). Retrieved May 31, 2023, from

<https://www.kaggle.com/datasets/ruqaiyaship/osha-accident-and-injury-data-1517>

Shekari, A. M. (2020). SAFETY MANAGEMENT SYSTEMS STANDARDS & GUIDELINES: A Comparative

Analysis. *Professional Safety*, 65(9), 32–37.

Sherratt, F., & Ivory, C. (2019). Managing “a little bit unsafe”: Complexity, construction safety and

situational self-organising. *Engineering, Construction and Architectural Management*, 26(11),

2519–2534. <https://doi.org/10.1108/ECAM-09-2018-0376>

Wang, H., [Link to external site](#), this link will open in a new window, Tian, Y., [Link to external site](#), this

[link will open in a new window](#), Yin, H., & [Link to external site](#), this link will open in a new

[window](#). (2021). Correlation Analysis of External Environment Risk Factors for High-Speed

Railway Derailment Based on Unstructured Data. *Journal of Advanced Transportation*, 2021.

<https://doi.org/10.1155/2021/6980617>

[REDACTED]

[REDACTED]

Yinghui, P., Tang, J., Biyang, M., Yifeng, Z., & Zhong, M. (2021). Toward data-driven solutions to

interactive dynamic influence diagrams. *Knowledge and Information Systems*, 63(9), 2431–

2453. <https://doi.org/10.1007/s10115-021-01600-5>
